Dobre Andreea Viktorya
AAB
2026

Laboratory 1

Requirement 1: Data Extraction & Cleaning

Overview:
Initial Sample: 9544 subjects
Cleaned Sample: 8756 subjects
Excluded Records: 788 incomplete records

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=== STEP 1 SUMMARY: DATA VALIDATION & CLEANING ===
Counting missing (NaN) values per target column:
RIAGENDR      0
BMXHT       775
BMXWT        99
dtype: int64
Total raw records: 9544
Total clean & valid records remaining: 8756
Excluded 788 incomplete records.
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Cleaned Dataset Structure
First 5 valid records: display complete paired observation for gender
Data Integrity: Confirms that all missing values were successfully removed and categorical values were mapped for analysis.

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Cleaned dataset sample (first 5 records):
 RIAGENDR  BMXHT  BMXWT Gender_Label
      1.0  184.5   94.8         Male
      1.0  171.4   90.4         Male
      1.0  170.1   83.4         Male
      2.0  160.9  109.8       Female
      2.0  164.9   55.2       Female
```

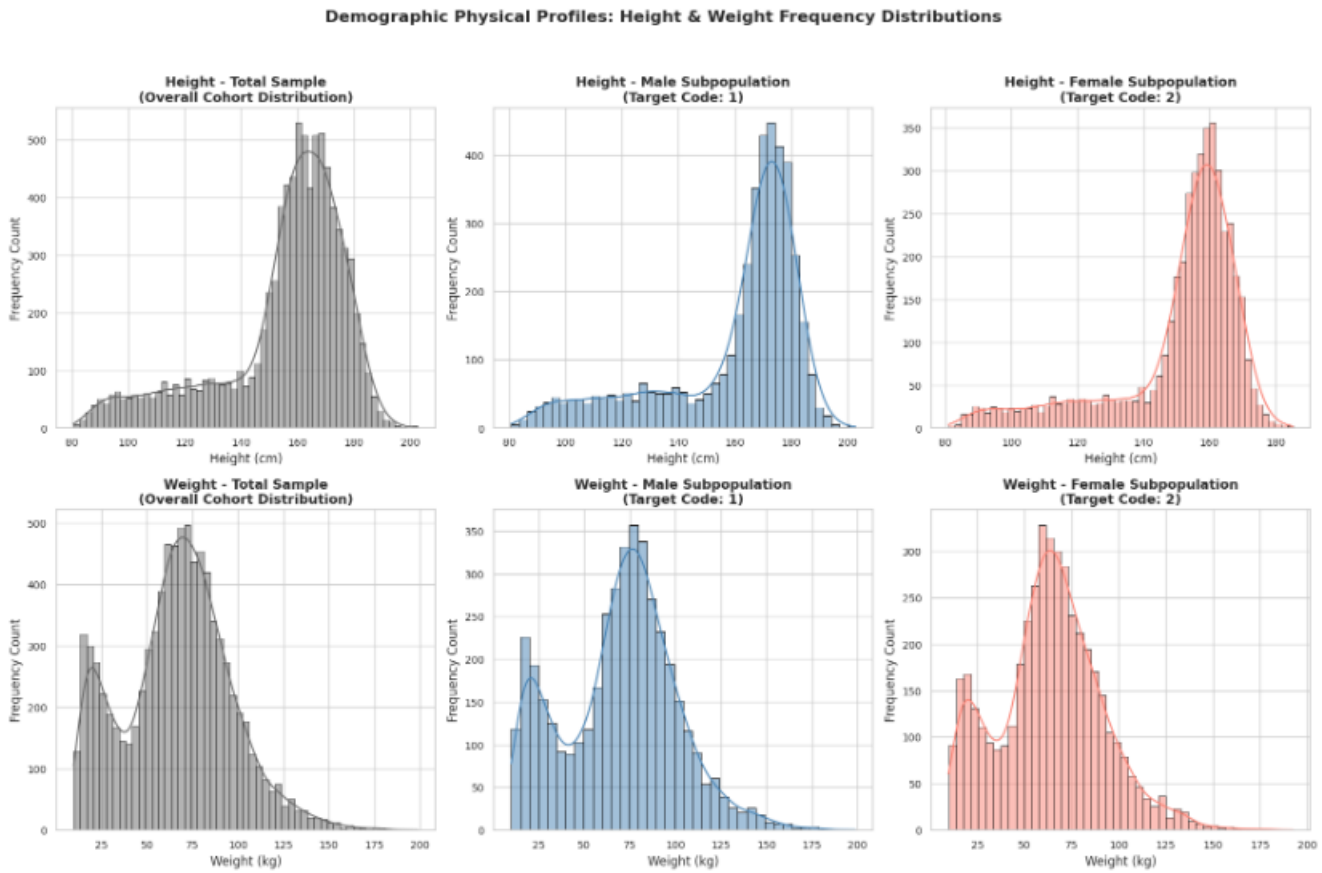
Requirement 2: Frequency Histograms & Distribution Analysis

Overview:
Frequency histograms with Kernel Denisty Estimation overlays were generated for both height anf body weight to analyze their distribution patterns across the entire cohort as well as separately for male and female.

Height Distribution Analysis
Total Sample: Display a major peak centered around 165-170 cm; Shows a secondary left tail between 80-170 cm, representing pediatric participants included in the NHANES survey.
Male Subpopulation: Follows an approximately bell-shaped distribution for adults, with the main peak centered around 175 cm.
Female Subpopulation: Exhibits a symmetric, bell-shaped distribution centered around 160-162 cm

Weight Distribution Analysis
Distribution Shape: Shows a strong right-skewed distribution across all three plots
Pediatric Subgroup: A district secondary peak is visible at 15-30 kg in all histograms, corresponding to children in the dataset.
Adult peak & high-weight tail: The main adult peak reaches 70-80 kg for males and 60-70 kg for females.

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Requirement 3: Descriptive Statistics & Bivariate Correlation Analysis

Descriptive Biometric Statistics:
Height:
Total Cohort: Mean = 155.08cm, Median = 161.10cm, Std Dev = 22.99cm
Male Subpopulation: Mean = 159.66cm, Median = 169.20cm, Std Dev = 25.09cm
Female Subpopulation: Mean = 150.68cm, Median = 156.90cm, Std Dev = 19.81cm
Distribution Pattern Standard deviation and variance are lower in females, indicating a more compact height distribution around the mean compared to males.

Weight:
Total Cohort: Mean = 66.60 kg, Median = 68.10kg, Std Dev = 29.87kg.
Male Subpopulation: Mean = 69.7kg, Median = 72.80kg, Std Dev = 31.52kg}
Female Subpopulation: Mean = 63.63kg, Median = 63.80kg, Std Dev = 27.87kg
Distribution Pattern: Weight exhibits significant variance across all groups, driven by the wide gap between minimum values and extreme maximum values.

Bivariate Correlation & Scatter Plot Analysis
Total Cohort Correlation: Demonstrates a strong positive linear relationship between standing height and body weight across the general population sample.
Male Subpopulation Correlation: Shows a very strong positive correlation, indicating that weight increases consistently with height in males.
Female Subpopulation Correlation: Reflects a strong positive correlation, slightly lower than in males, likely due to higher biological variation in body composition (body fat vs. muscle mass).
Scatter Plot Dynamics: The linear regression lines across all subplots confirm the positive trend. Scatter dispersion increases notably at higher height intervals, showing greater weight variability among taller adults.

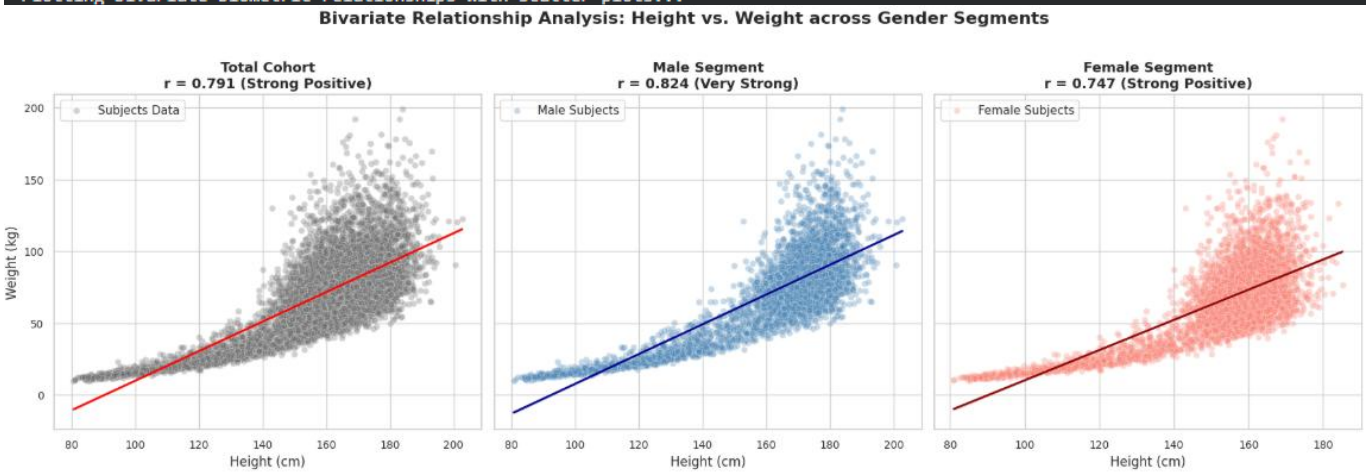
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=== STEP 3: SUMMARY OF DESCRIPTIVE BIOMETRIC STATISTICS ===
        Cohort Biometric Variable  Sample Size (N)    Mean  Variance  Std Dev  Minimum  Q1 (25%)  Median (50%)  Q3 (75%)  Maximum
  Total Cohort        Height (cm)             8756 155.08    528.73    22.99     80.7    148.90         161.1     170.4    202.7
  Total Cohort        Weight (kg)             8756  66.60    891.95    29.87      9.8     47.80          68.1      85.2    198.9
  Male Segment        Height (cm)             4286 159.66    629.70    25.09     80.7    152.62         169.2     176.2    202.7
  Male Segment        Weight (kg)             4286  69.71    993.50    31.52      9.9     48.52          72.8      89.4    198.9
Female Segment        Height (cm)             4470 150.68    392.56    19.81     81.0    148.00         156.9     162.9    185.2
Female Segment        Weight (kg)             4470  63.63    776.68    27.87      9.8     47.32          63.8      80.5    192.3

Computing Pearson Correlation Coefficients (r)...

=== PEARSON CORRELATION METRIC RESULTS ===
Overall Cohort Pearson Correlation Coefficient (r) : 0.7911
Male Subpopulation Pearson Correlation Coefficient (r): 0.8245
Female Subpopulation Pearson Correlation Coefficient (r): 0.7468

Plotting bivariate biometric relationships with scatter plots...
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The scatter plots display the bivariate relationship between standing height and body weight across the total cohort and within gender-specific subgroups. In all three plots, the fitted linear regression lines confirm a strong positive correlation, indicating that body weight increases predictably with height.

For the overall sample, the tight alignment of data points at lower height and weight levels reflects the pediatric growth trajectory, where physical growth occurs concurrently in both dimensions. In the male subpopulation, the relationship is particularly strong, showing a steady upward trend. The female subpopulation displays a slightly lower correlation, which is attributable to greater variations in body composition, such as adiposity and muscle density.